

RISK ASSESSMENT REPORT

Financial Anomaly & Risk Scoring Engine — Generated on August 07, 2026 at 00:01:57

Executive Summary

A batch prediction run was executed using the **best_model.joblib** classifier. A total of **10,000** transactions were analyzed. Out of these, the engine flagged **187 (1.87%)** transactions as **High Risk** anomalies. The average risk score across the dataset was **1.9/100**, and the model's confidence level on flagged anomalies averaged **95.9%**.

Total Transactions	10,000	High-Risk Flags	187 (1.87%)
Total Volume	\$3,208,239.44	Avg Risk Score	1.9 / 100
Unique Customers	500	Anomaly Confidence	95.9%

Dataset & Context Profile

Metric	Value Description
Earliest Transaction	2026-06-26 22:40:37
Latest Transaction	2026-07-27 09:43:44
Average Transaction Size	\$320.82
Unique Customers Active	500 customer profiles evaluated

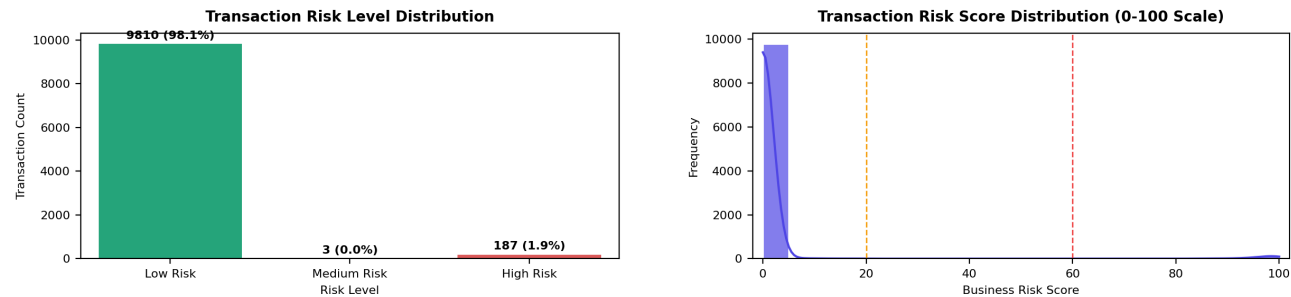
Model Performance Profile

The active classification model is **best_model.joblib**. During offline validation, this model demonstrated an overall Accuracy of **99.90%**, Precision of **97.30%** (low false alarm rate), and Recall of **97.30%** (anomaly catch rate).

Evaluation Metric	Active Model Baseline	Current Run Actual
Accuracy	99.90%	99.99%
Precision (False Positive Mitigation)	97.30%	99.47%
Recall (Fraud Anomaly Capture Rate)	97.30%	100.00%
F1-Score (Balanced Performance)	97.30%	99.73%

Risk Analytics & Visualizations

Below are the analytical plots mapping the transaction risk profiles for the current batch prediction run. The left chart visualizes counts and percentages by risk tiers, while the right chart displays the score distributions.



Risk Category	Risk Score Range	Transaction Count	Proportion
Low Risk	0.0 - 20.0	9,810	98.10%
Medium Risk	20.1 - 60.0	3	0.03%
High Risk	60.1 - 100.0	187	1.87%

Business Insights & Operational Recommendations

Based on the analysis of risk levels, average risk score, and model prediction confidence, the system has generated the following operational recommendations for risk mitigation:

[INFO] Standard Automated Processing

High Risk transactions are within normal operating limits (1.87%). We recommend: (1) Continuing with standard automated processing. (2) Standard daily reconciliation report audits. (3) No immediate operational interventions.

[INFO] Systemic Risk Stable

The average transaction risk score is stable at 1.9/100, indicating healthy transactional behavior across the network.

[INFO] High Confidence Detections

The machine learning model is operating with high certainty (average anomaly confidence of 95.9%). The automated decisions and risk scoring can be highly trusted for system actions.